KISII UNIVERSITY

UNIVERSITY EXAMINATIONS

THIRD YEAR EXAMINATION FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE IN APPLIED COMPUTER SCIENCE

SECOND SEMESTER 2023/2024

[JAN – APRIL, 2024]

COMP 302: DESIGN AND ANALYSIS OF ALGORITHMS

STREAM: Y3 S2 **TIME: 2 HOURS**

DAY: THURSDAY, 12:00 - 2:00 P.M. **DATE: 04/04/2024**

INSTRUCTIONS

1. ***Do not write anything on this question paper.***
2. ***Answer question ONE (Compulsory) and any other TWO questions.***

QUESTION ONE (30 MARKS)

a) Explain what an algorithm is and outline three properties of algorithms (3 marks)

b) Construct an algorithm to display the GCD of two integers m and n where m>n (5 marks)

c) Using the GCD algorithm in Question 1b, explain the basic characteristics of an algorithm (5 marks)

d) Write an algorithm for the binary search method (6 marks)

e) Consider the following Java method:

```
int a[5];
int mystry(void)
{
```

```
            int i;
            a[0] = a[1]=1;
        for(i = 2; i< 10; i++)
            a[i] = a[i -1] + a[i - 2];

            return 0;

        }
```

i. Write down the contents of the array a after execution of the mystry method above (6 marks)

f) Compare and contrast *divide and conquer* and *greedy* algorithm design strategies. (5 marks)

QUESTION TWO (20 MARKS)

a) Identify an appropriate sorting algorithm and use it to answer the following questions;

   i. Determine the execution time equation of the algorithm (5 marks)

   ii. Determine the Big O and Theta notations. (5 marks)

   iii. Determine the nature of the data sets that make the algorithm to have best case, worst case and average case. (5 marks)

b) Given that $f(n)=1/2n^2 -3n$ prove that $f(n)=O(n^2$ (5 marks)

QUESTION THREE (20 MARKS)

a) Define the term "spanning tree" (4 marks)

b) Describe divide and conquer method, write an appropriate control abstraction (6 marks)

c) Explain the Kruskal's algorithm for finding the minimum cost of a spanning tree (6 marks)

d) Illustrate the workings of the Greedy algorithm design technique (4 marks)

QUESTION FOUR (20 MARKS)

a) Provide any three application areas of the Travelling Sales Man problem (3 marks)

b) Construct the pseudocode for insertion sort given the number of elements is small (7 marks)

c) Write the Dijkstra's algorithm to find the solution for a single source shortest path problem using the Greedy Technique (10 marks)

QUESTION FIVE (20 MARKS)

a) Explain the two major factors to consider when computing the complexity of an algorithm (2 marks)

b) Discuss any TWO criteria of analyzing the running time of an algorithm. State one advantage of each. (8 marks)

c) Construct a spanning tree from the connected graph given using Prim's technique (10 marks)